

An augmented reality based Virtual dressing room using Haarcascades Classifier

Nauman Zafar Hashmi¹, Aun Irtaza¹, Wakeel Ahmed¹, Nudrat Nida²

¹Department of Computer Science, University of Engineering and Technology, Taxila, Pakistan

²Department of Computer Science, Air University, Aerospace and Aviation campus Kamra

Correspondence Author: Nauman.hashmi@students.uettaxila.edu.pk

Abstract— Virtual dressing rooms are modern way of shopping where customers experience realistic dress fitting and order online. As trying different dresses at shopping centers is very time consuming and customers are not sure how the dress looks on them. Therefore, there is a need to identify efficient methods to improve virtual dressing rooms and increase e-commerce businesses. All the existing virtual dressing room methods required Kinect sensors for virtual dress fitting and Kinect sensors are expensive. Our motivation is to create such virtual dressing room that overcome the constraint of Kinect sensor input. We present a dynamic virtual dressing room based on web cam to acquire input video and allow user to experience virtual dress fitting. Our proposed model comprised of following steps including face, skin, lower body detection and dynamic distance estimation for precise virtual reality. We have evaluated our algorithm on 50 subjects using 10 dresses.

Keywords— *Augmented Reality, Virtual Dressing room, Image Processing, Virtual Reality, E-commerce.*

I. INTRODUCTION

Virtual Dressing Room is an environment which provides the facilities to the customer to try their dresses virtually. As online customers prefer to buy products online and also want to experience the appearance of dresses. Moreover, lot of customers on special occasions desire to have virtual dressing rooms to save time and fatigue. Due to large number of customers in boutiques, it's been a difficult task for the customers to buy the products and try their favorite dresses in a try room.

Digital technologies have been playing an important role over the past years in fashion and business industry. It has great impact on our everyday life. Because of the versatile

usage of augmented reality and virtual reality techniques, information technology has become a new model for fashion industry. Virtual try rooms of dresses has received much attention recently due to its commercial potential. The evolution of this new technology in fashion design industry will lead to customer satisfaction which can be economically apply to all virtual shopping stores. Currently, there are huge advance in bringing VR and AR research from laboratories to industry, especially in game markets. Benefit from hardware upgrades and blooming in artificial intelligence, there is still huge potential for VR and AR technology in a wide range of areas. Li *et al.* (2019)

The application in business oriented and follow all SMART rules of the business organization (N, 2000). These Rules determines the management strategy for business and our product completely satisfies with specificity, measurability and achievability of the product. It is more realistic and time based. Introductory work has been done on correlating the design item on human body virtually in such a way that user stand in front of the automated embedded system mirror. The system virtually maps the fashion fabric to body object resulting a realistic display. The limitation of their work is that the system doesn't map the dresses accurately along the body object. For more realistic display for user's satisfaction, it is important that the system should fit the dresses precisely. "I Fashion Clothing System" and "Magic Mirror coordination System", introduce virtual dressing rooms using Kinects and RFID tracking to identify customers.

Earlier, Computer Aided Design (CAD) was used for designing 3D models for dresses and use them as an asset

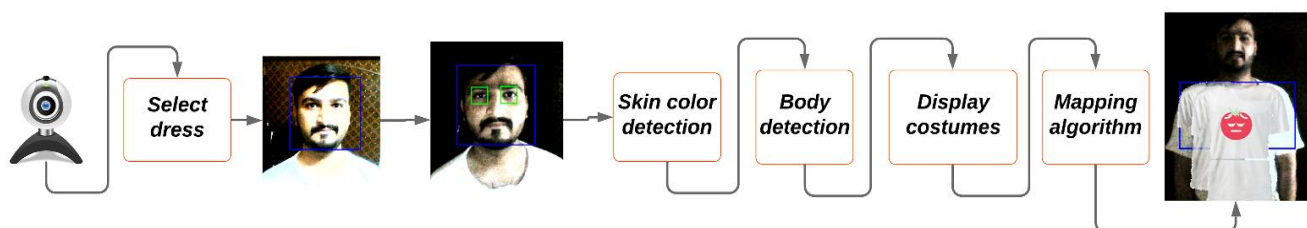


Figure 1 System Architecture

with Kinect sensor. Yaun *et al.* (2013).). (Kim and Cheeyong, 2015) proposed magic mirror coordination system which identify the subject, analyze the body data digitally. User has to place the desired hair styles, makeups and garments manually by clicking on the button and select their preferred style. Masri *et al.* (2019) described a new augmented reality concept for dressing rooms using Kinect sensor. Their methodology includes 2D clothing and the study aims a main focus on the extraction of body features by indulging Kinect sensor directly on the body features, which fits in the 2D garments. Fathima *et al.* (2019) describe virtual try room using visual animations “ShopAR”. Bhalekar *et al.* (2017) introduced a technique which gives a perfect look to a skeleton by observing different characteristics e.g. hair color, height, skin tone etc. using camera, motion and light sensors. Zaware *et al.* (2018) suggested a model for virtual dressing room based on smart phone using Euclidian distance formula has been used in algorithm to measure the body of customer. Virnes *et al.* (2018) proposed techniques for detecting outfit number, size, color etc. in dressing room. They also introduced labels and price tags along dress color, size and type for smart dressing room or electronic dressing room. Feng *et al.* (2019) introduced finger tracking technique using augmented reality which enable the customer to customize their dresses in virtual fitting room. RGBD sensors and 3D model has been used. Chopra (2019) introduced artificial intelligence tools like unity cross platform in virtual rooms using augmented reality which fascinated youngsters to avail this novel AI Vrooms and shop online with no fear of doubt. Rosy *et al.* (2019) focused on technological advancement of fashion industry in few years using augmented reality and virtual reality. Research has been conducted to check the “perceived usefulness” and “ease of use” of augmented reality and virtual reality an also its impact and benefits in future. The study gives a complete and comprehensive comparison of usefulness among customers and results constitutes more in favor of easiness.

All the existing techniques are based on Kinect sensor input and Kinect cameras are expensive. In order to introduce cost effective technique for virtual dressing room, we propose a scheme using Haarcascades algorithm with skin detection algorithm to precisely fit the dresses on customers. After the detection, dresses will be mapped on the body using mapping algorithm. This algorithm predicts the locus from which the read originates and gives the best possible fit of the dress on the body. Mapping algorithm determines the general constraint satisfaction problem; therefore, the outcomes would be finest by using mapping algorithm. Size of the dresses will be adjusted dynamically through Euclidean distance. It will adjust the size of the material with the measurement of body through Euclidean distance technique. Consequently, Subject will get the desired result through camera and can patterned their dresses with affluence by the help of our product. Our contributions are summarized as follows: (i) we introduce a new scheme for virtual try room using ordinary camera. (ii) our system overcome the problem of environmental constraints including low contrast, view point variation, illumination changes. (iii) our system is computationally inexpensive.

II. PROPOSED METHOD

The proposed approach acquires dynamic video using web cam then Haarcascades classifier is being employed to detect different parts of body including head, face, eyes and lower body followed by skin detection algorithm. Then distance is estimated to adjust the operational distance between the camera and the subject. The model of the dress is then scaled to adjust on subject. The video of the subjects allows the superimposition of dress model on real time as shown in Figure 1.

The proposed method is discussed in following sub sections.

A. FACE AND LOWER BODY DETECTION

Haarcascades classifier is used for face detection and lower body detection. Face is an essential entity for detecting that subject is present within the operational frame. Haar uses different type of contrast to detect the parts easily. The rectangles shown in Figure 2 represent the group of pixels. These pixels will help to detect the body part quickly. Face and body detection uses haar feature.

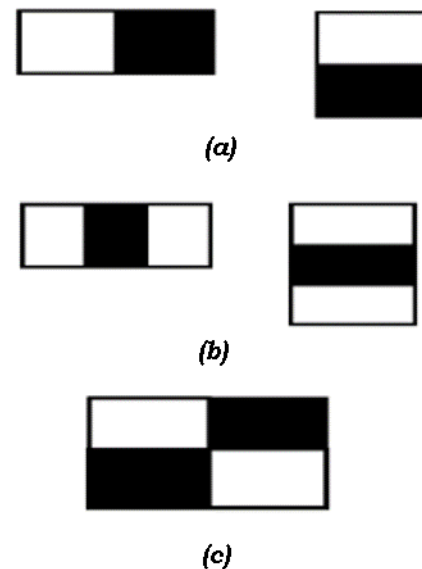


Figure2: (a) Edge Features (b) Line Features (c) Four Square Features

Haar classifiers are used to perform the lower body detection. Detection of the humans in a video that is moving; features like detecting eye, face and body are performed by the haar classifiers. The negative and positive samples of images are inculcated by the help of haar classifiers and saved as .xml files.

Haarcascade_upperbody,

haarcascade_lowerbody,haarcascade_fullbody,

haarcascade_frontalface etc. are those cascades used in

OpenCV to detect a human. Region of Interest (ROI) is used by the haar. It is based on the “Haar Wavelet Technique” for verbally analyzing pixels in squares. This is called “training data” which uses machine learning techniques to achieve higher degrees. To estimate its properties the concept of “integral image” has been used. The Ad boost learning algorithm is being used by Haarcascades, which selects a number of key features from a large set to give the classifier

efficient results. In order to detect the lower body by the help of haar classifier, a flow chart is shown in Figure 3.

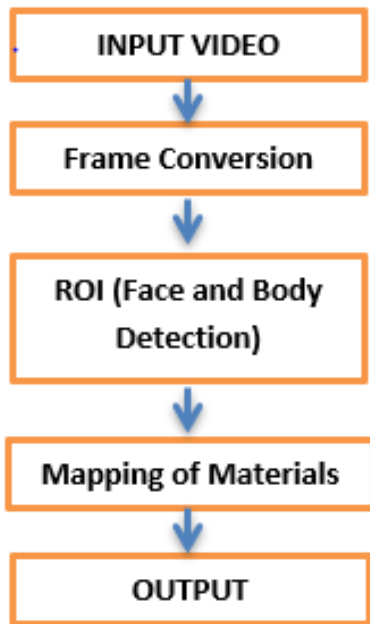


Figure 3: Flowchart for Face and Body Detection

It will return location of face in term of width and height and pixel coordinates as in Figure 4.

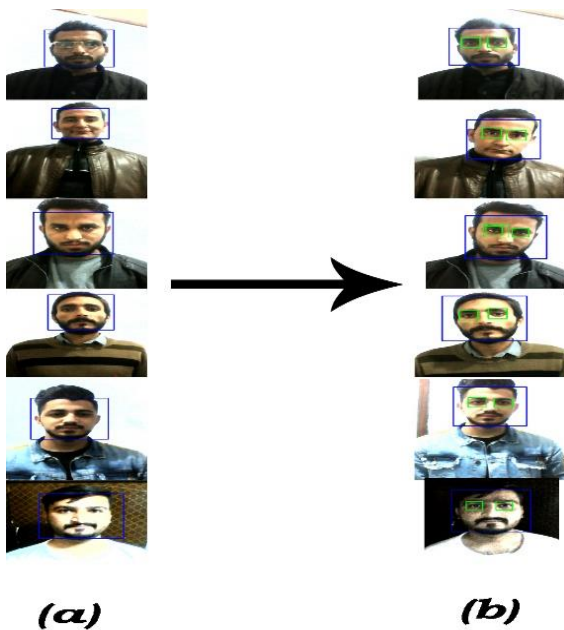


Figure 4: (a) Face Detection (b) Region of Interest (ROI)

B. SKIN COLOR DETECTION ALGORITHM

Then the detected faces and lower body parts are further processed to identify skin pixels. Skin color detection algorithm is need to use here in order to maintain the novelty

of product. By using this algorithm, the undetected body parts would get fixed into the desired dress with perfection of net-zero difference. For skin detection algorithm, we transform RGB image to YCbCr color space to discriminate chrominance from intensity channel. Y indicates luminance information, Cband Cr indicate chrominance information. Cb and Cr represent blue and red color differenced with respect to reference value respectively. Y includes range of 8 bit that is 16-255, Cb and Cr are blue and red components having range 16-240. The equation of transformation of RGB to YCbCr is given in Equation A,

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 16 \\ 128 \\ 128 \end{bmatrix} + \begin{bmatrix} 65.481 & 128.553 & 24.966 \\ -37.797 & -74.203 & 112 \\ 112 & -93.786 & -18.214 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} \quad (A)$$

The conforming skin pixel is in the range which is stated below:

$$y > 80 \quad (i)$$

$$85 < Cb < 135 \quad (ii)$$

$$135 < Cr < 180 \quad (iii)$$

Where Y, Cb, Cr = [0, 255]

The skin color varies from person to person; therefore, a threshold range is set to cover all types of humans like Asians, Europeans etc. A histogram technique is used to segment the skin color pixels within the video frame. Histogram analysis will result in discrimination of skin pixels using optimal threshold. The threshold value will give the classifier or filter stage. The pixel number with human skin is counted by the Skin Pixel Quantifier to estimate the percentage of skin pixels and segment out it.

Bounding box on a subject which shows on real time is the region of interest. The ROI image is obtained through detection using Haarclassifier and bounding box shown on the body. Dresses are mapped on the subject through mapping algorithm and this will be applied on body after proper detection. This mapping will give the desired result according to the measurement of the body of subject.

Table1. Show the readings of body detection measured by Haar classifiers.

Haar-cascade classifier	Label	Stage	Size
Haarcascade_frontalface_default	HD	25	24 × 24
Haarcascade_frontface_1	HT	22	22 × 22
Haarcascade_frontface_2	HT2	21	22 × 22
Haarcascade_frontface_alternative_tree	HTR	48	25 × 25
Haarcascade_profile	HPF	30	25 × 25

Table 1. Haar-cascade classifiers of different sizes

C. DISTANCE CLASSIFIER TECHNIQUE

The Euclidean distance is used to estimate the dynamic distance between the camera and the subject. When the

distance between the subject and camera is huge then it will automatically rescale to adjust the dress on body precisely. Hemant *et al.* (2017)

The following shows the Euclidian distance formula:

$$\Delta s((x, y), (a, b)) = \sqrt{(x - a)^2 + (y - b)^2} \quad (1)$$

Where x and y are camera centroid coordinates and a and b are the subjects centroid coordinates respectively. Δs represents the distance. The Euclidian distance used to estimate the length of a distance between two points either in the plane segment or 3-dimension segment. This technique is assumed as the most précised and reliable way to estimate distance between two points. We have used Euclidean distance formula based on Pythagorean formula to estimate the distance. We have defined four points x, y, a, and b respectively. The Euclidean distance can be computed by:

$$\sqrt{(x - a)^2 + (y - b)^2}.$$

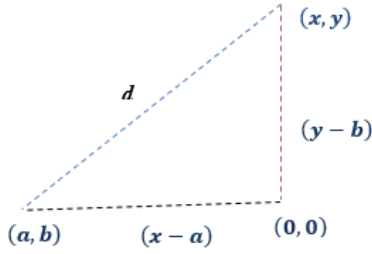


Figure 5: Euclidean Distance between Two Points

D. MAPPING ALGORITHM

Mapping algorithm is a form of intense technology which understands the physical world through feature points. Mapping algorithm used in proposed method help for applications based on augmented reality to recognize and identify 3D objects. Along with it, it instantly tracks the object and make the overlay digital interactive augmentation possible. In our proposed method the device targets on the subject, on its 3D position and calculate the spatial distance between object and classifies at multiple key points.

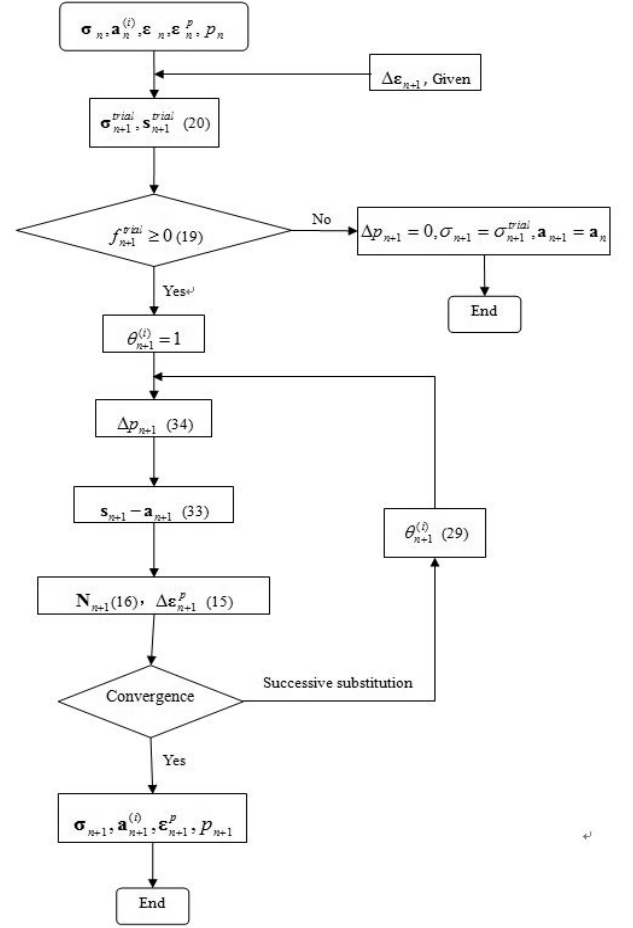


Figure 6: Flowchart of Mapping Algorithm for Rate-independent model

III. RESULTS

We have evaluated our proposed scheme on 50 subjects with 10 dresses. The dynamic video frames are captured through camera and our algorithm precisely fit the dresses on subjects. Our input image transformed from RGB colors to YCbCr and formerly using mapping algorithm threshold would be calculated. Which gives a quantified skin pixel and desired output to the customers.

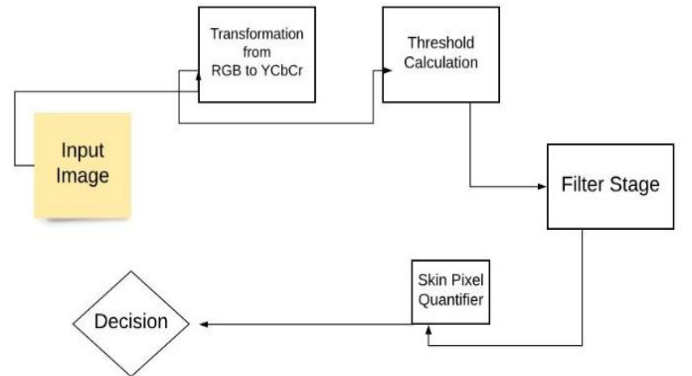


Figure 7: Processing- Input to Output

IV. COMPARISON.

The existing systems used Kinect sensors. One of the existing systems “Virtual fitting room possible by AR application” use ordinary camera. This system was based on the static augmented reality in which costume display on the screen by default and subject have to adjust itself accordingly. And it does not give precise results as there is no detection of the body.

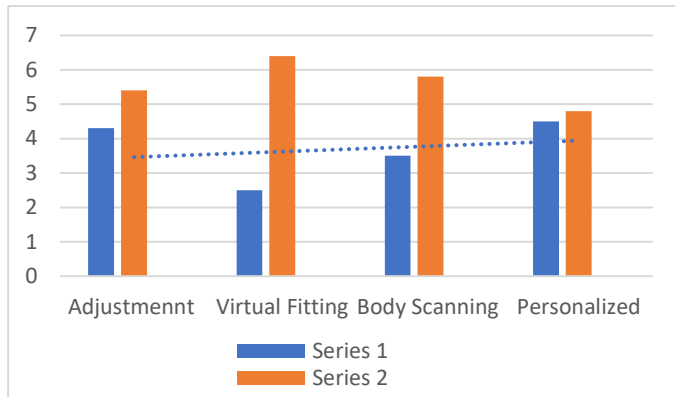


Figure 8: Comparison of features with AR model

Size of the body was not measured and costumes did not map precisely on the body.

In the figure 8, the graphs describe the comparison of already standing AR models with our model, after critically analyzing the results of several AR models the comparability increased in virtual fitting and body scanning feature of our model. That is because the mapping algorithm perfectly maps the body and usage of go-ahead skin detection algorithm for all skin colors is original. Therefore, the results of comparability are clearer.

Features	AR models (static AR)	Proposed Model
Adjustment	4.0	5.5
Virtual fitting	2.0	6.2
Body scanning	3.5	5.8
Personalized	4.0	5.0

Table 2: Comparison with AR application fitting using static augmented reality

The rate of satisfaction over 10 tries of both general AR models and proposed model gives the results that, virtual fitting and body scanning is comparatively higher in our proposed model.

Our model works on dynamic augmented reality. Detection of the body either upper or lower take place and size of the dress adjust on the body according to the size of subject. Our model is useful and beneficial for every individual either men or women because our model is highly inexpensive and more affordable system in this expensive and affluent ambience. The digital services afford by our proposed model comprises of 3D body fitting, magic mirror, mapping subject via mapping algorithm and 3D body scanning. Eirini *et al.* (2019)

Digital services on different e-commerce sites	Proposed model and Services	Compatibility: Proposed model V/S Static AR applications
3D virtual body representations that consumers can access online through mobile phones.	3D virtual body fitting, mapping subject, and 3D body scanning.	High

Table 3: Comparison of our model with the existing application present on the internet

If we compare with the magic mirror coordination where when a subject is placed in front of magic mirror and as a result dress will be imagined on the subject. Magic mirror mainly works on feature reduction. The comparison of feature extraction and feature reduction is shown in the graph which is as follows:



Figure 9: Comparison with Magic Mirror System.

Our model is different from others in correspondence with the whole-body extraction algorithm which detects the skin color of every type and its novel in its own way. Whereas other products run on only skin detection in general and allows users to virtually try their dresses. Mapping algorithm gives the overall outline of the body and map the dress accurately on the detected body parts.

V. CONCLUSION

Shopping centers and malls use real time virtual dressing room, as trying dresses in the shopping centers is crowded and privacy issues. Our motivation is to create such cooperative and highly satisfactory virtual dressing room. Our proposed model is comprised of following steps including face, skin, lower body detection and Euclidian distance followed by dress mapping on subject body.

We introduce computationally inexpensive technique using web cam input.

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